

AI Adoption Prediction Across U.S. States

A Decision-Support System for Digital Readiness & Policy Planning



The Problem

AI adoption is uneven across U.S. states

We don't know who is ready, who is behind, or why.

- Economic competitiveness at stake
- Workforce development gaps
- Small business growth limited
- Public investment decisions unclear

Leaders currently guess → this system makes it data-driven

Policymakers

Economic Teams

Investors

Digital Programs

How ML Helps

Combines Data

Census, FCC, SBA, and education data merged

Predicts Readiness

Each state's AI readiness probability calculated

Explains Drivers

Identifies what influences readiness most

Runs Scenarios

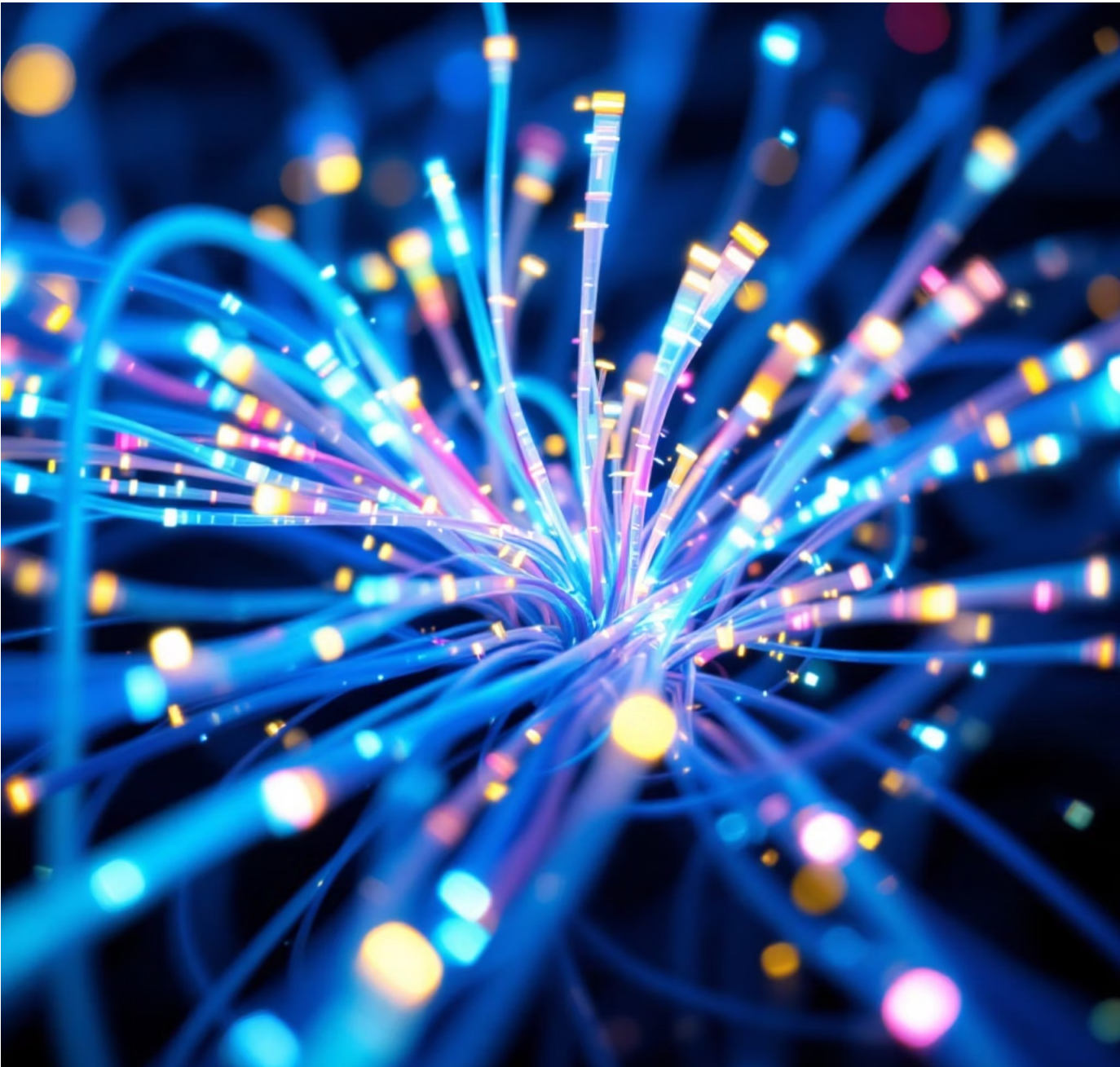
What-if analysis: +5% broadband impact readiness +12% Mississippi

A decision-support model that guides smarter AI planning and investment



Key Insight

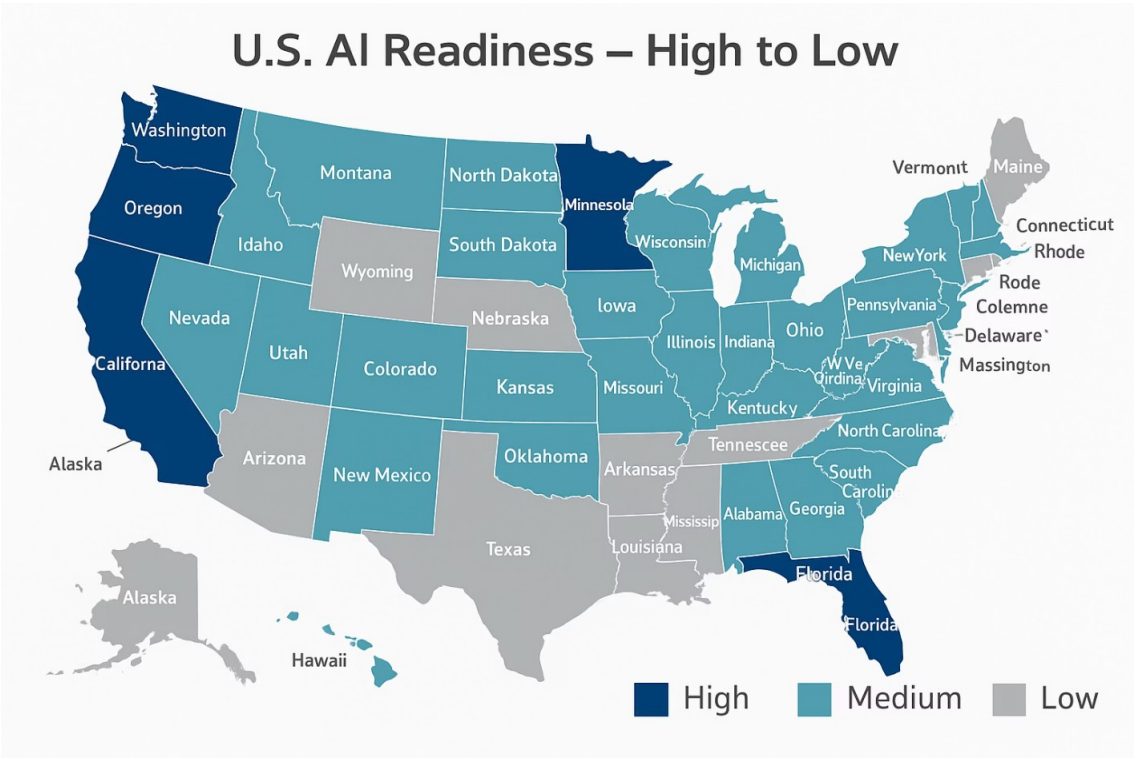
Broadband is the #1 Driver



Connectivity predicts readiness more than anything else

Digital divide → AI divide

States with stronger broadband infrastructure show significantly higher AI adoption potential.



Key Insight

Human Capital

Bachelor's and STEM rates heavily influence readiness
Skills + Infrastructure = AI Adoption

SME Readiness Gap

Small businesses lack digital capability despite interest
SMEs must adopt AI for statewide transformation

Income Interaction

Income boosts readiness only when paired with education & broadband
Economic strength $\neq$ AI readiness without digital foundations

Innovation Density

States with strong tech hubs perform better
Innovation accelerates AI, but cannot replace fundamentals

Regional Divide

Northeast & West Coast lead; South & Mountain West lag
Structural gaps create long-term AI inequalities



Critical Challenges Identified



Small Businesses Lag

SMEs want AI but lack access



Policy Gap

Need training and policy support



Ethical Risk Identified

Readiness scores can be misused



Governance Actions

Model card and governance matrix



Ethical Guidelines

Implement principles and controls



ML Lesson Learned

Simpler models, strong features

The Big "So What?"

 **AI adoption depends on three pillars:**



Broadband

Infrastructure foundation



Skills

Human capital development



Responsible Deployment

Ethical governance

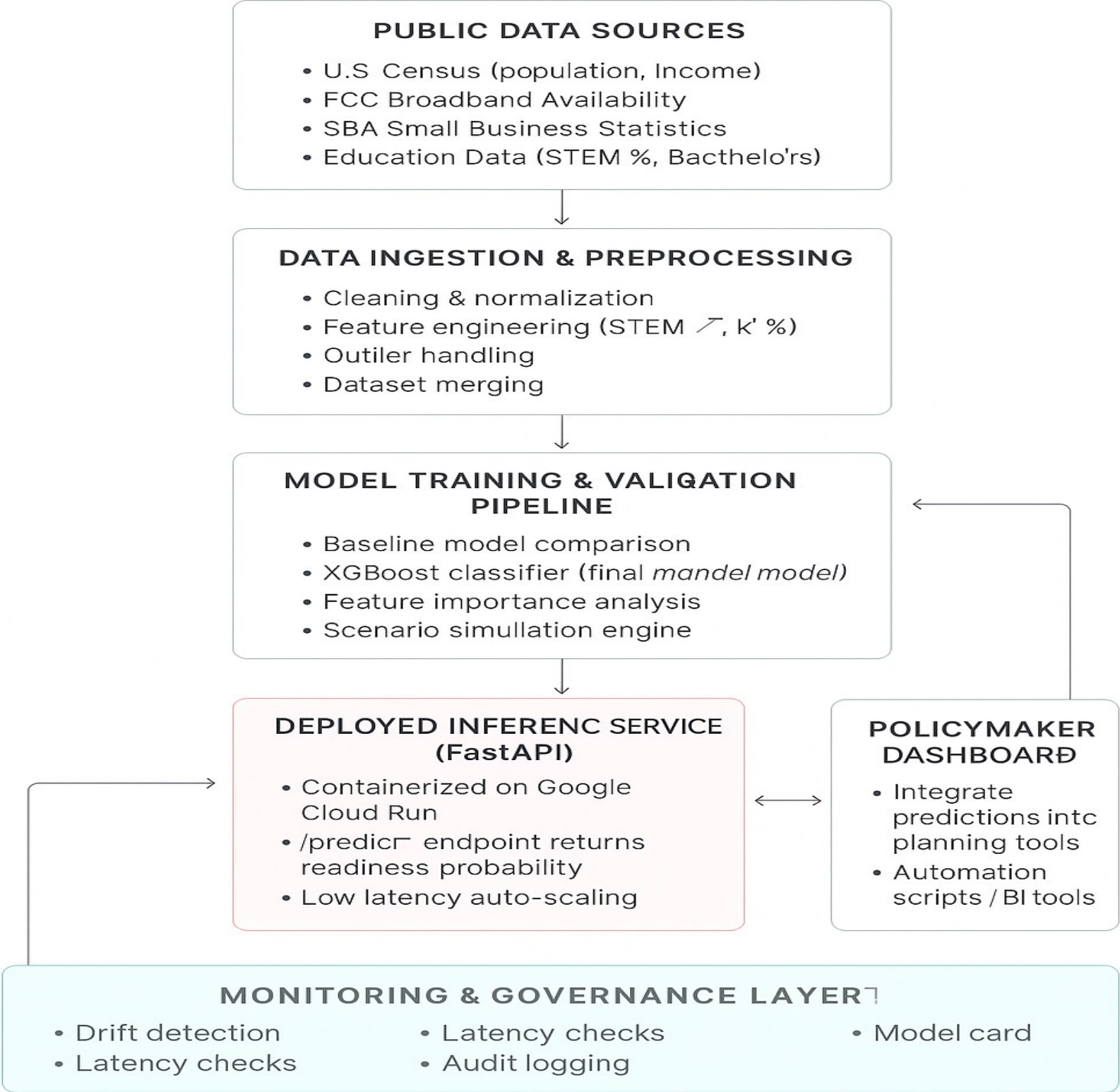
Not just technology — it's about connectivity, capability, and accountability.



End-to-End System Architecture

- **Public data** (Census, FCC, SBA, Education) feeds into a unified dataset.
- **Preprocessing pipeline** cleans, normalizes, and engineers features.
- **XGBoost model** is trained, validated, and used for scenario simulations.
- **FastAPI service** deploys the model as a real-time “/predict” endpoint.
- **Dashboard** gives policymakers readiness scores and what-if insights.
- **Monitoring layer** tracks drift, latency, logs, and ensures ethical use.

End-to-End System Architecture





System Components

Data & Model

01

Public Data Integration

Income, population, broadband, business stats, education

02

XGBoost Selection

Chosen after model comparison for accuracy

03

Scenario Analysis

Improvement simulations and predictions

Deployment & Governance

01

Production API

Low-latency, autoscaling FastAPI service

02

User Access

Dashboard for policymakers, API for apps

03

Responsible AI

Drift monitoring, latency tracking, audit trails



Impact

A practical, production-style ML pipeline that turns public data into actionable AI readiness insights

50

States Analyzed

Comprehensive U.S. coverage

4

Data Sources

Integrated for complete picture

1

Unified System

Data to deployment pipeline

Where investment pays off most — guiding smarter AI planning across America